General Realtivity Excercise 1

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1 Warmup

1.1 The Metric is symmetric

First of all, the metric tensor must be a symmetric one. This can be seen from the symmetry of the definition. Assuming $g_{ij} \neq g_{ji}$, we get

$$ds^2 = g_{ij}dx^i dx^j \neq g_{ji}dx^i dx^j = ds^2 \tag{1}$$

Which of course can't be the case, hense the metric is always symmetric.

1.2 A non-uniform metric

The most simple example of a non-uniform metric is the metric of the 2d Eucledean space in polar coordinates:

$$g^{ij} = \begin{pmatrix} 1 & \\ & r^2 \end{pmatrix} \tag{2}$$

1.3 Inveriance of ds^2 under change of coordinates

Let there be two sets of coordinates x^μ and x'^μ, and the metric is defined for the first set of coordinates such that

$$ds^2 = g_{\mu\nu}dx^\mu dx^\nu \tag{3}$$

The transformation between the two sets of coordinates is given by the Jacobian:

$$\begin{gathered} J^\mu_\nu = \frac{\partial x'^\mu}{\partial x^\nu} \qquad \left(J^{-1}\right)^\mu_\nu = \frac{\partial x^\mu}{\partial x'^\nu} \\ dx'^\mu = J^\mu_\nu dx^\nu \\ g'_{\mu\nu} = \left(J^{-1}\right)^\sigma_\mu \left(J^{-1}\right)^\tau_\nu g_{\sigma\tau} \end{gathered} \tag{4}$$

So, using the property of the Jaccobian

$$J^\mu_\alpha \left(J^{-1}\right)^\beta_\mu = \frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x^\beta}{\partial x'^\mu} = \frac{\partial x^\beta}{\partial x^\alpha} = \delta^\beta_\alpha \tag{5}$$

The norm in the x'^μ coordinate system, is

$$ds'^2 = g'_{\mu\nu}dx'^\mu dx'^\nu = \left(J^{-1}\right)^\sigma_\mu \left(J^{-1}\right)^\tau_\nu g_{\sigma\tau} J^\mu_\alpha dx^\alpha J^\nu_\beta dx^\beta = \delta^\sigma_\alpha \delta^\tau_\beta g_{\sigma\tau} dx^\alpha dx^\beta = g_{\sigma\tau}dx^\sigma dx^\tau = ds^2 \tag{6}$$

1.4 Another example for the usage of a non-measurable variable

A good example for that would come from the ladder operators in QM. In addition to the x and p operators, we create another form of looking at the data, which is unobservable but useful in certain situations.

2 Properties of the Lorentz Transformation

2.1 The Lorentz Group

2.1.1 Definition

A Lorentz transformation Λ is a linear transformation that preserves the norm $ds^2 = c^2dt^2 - d\vec{x}^2$.

2.1.2 Multiplication

Let Λ_μ^ν and Λ'^ν_μ be two Lorentz transofrmations. Their product is also linear, since the product of the two linear transformation is linear.
Now, let v^μ be a vector and ds^2 its squared norm. By the definition, the vector $v'^\mu = \Lambda_n u^\mu v^\nu$ has the same norm ds^2, and again, by the definition, the vector $v''^\mu = \Lambda'^\mu_\nu \Lambda^\nu_\sigma v^\sigma$ has the same norm as v'^μ and v^μ. This means that the product of the two transformations $\Lambda' \circ \Lambda$ also preserves the norm and therefore a Lorentz transformation.